

ORANGE, Australia

WMO: 957260

Lat: 33.3814S

Lon: 149.1269E

Elev: 3112

StdP: 13.12

Time Zone: 10.00 (AUS)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.7	29.8	19.6	16.5	55.2	23.0	19.4	48.4	27.4	43.7	24.5	42.8	3.5	190	0.575

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.8	89.2	62.8	85.5	61.9	82.4	61.2	67.6	78.8	66.1	77.1	64.7	74.9	12.5	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	102.5	70.2	63.0	96.8	69.2	61.4	91.2	68.3	33.9	78.3	32.5	77.0	31.4	75.0	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.4	20.9	18.8	DB	23.3	93.7	1.9	4.6	21.9	97.1	20.8	99.8	19.8	102.3	18.4	105.7	(4)
(5)			WB	22.4	70.2	1.8	1.5	21.1	71.3	20.1	72.2	19.1	73.0	17.8	74.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.8	68.2	66.0	61.2	54.2	46.7	42.8	41.2	42.7	48.1	53.1	59.0	63.6	(6)	
	DBStd	10.69	6.33	5.88	5.33	5.26	4.65	4.08	3.89	4.09	5.50	6.22	6.72	6.16	(7)	
	HDD50	1013	0	0	1	20	123	218	274	230	101	38	7	1	(8)	
	HDD65	4403	40	51	141	326	569	667	738	691	507	369	203	101	(9)	
	CDD50	2406	565	447	349	145	19	1	3	44	135	276	421		(10)	
	CDD65	321	140	78	23	1	0	0	0	0	1	22	56		(11)	
	CDH74	3868	1610	818	278	26	0	0	0	1	37	345	753		(12)	
	CDH80	1227	611	268	49	1	0	0	0	0	1	91	206		(13)	
(14) Wind	WSAvg	9.2	9.5	9.2	9.1	7.5	8.3	9.2	9.7	9.9	9.9	9.5	9.4	9.3	(14)	
(15) Precipitation	PrecAvg	35.1	3.5	3.2	2.0	2.2	2.6	2.6	3.5	3.8	3.1	3.2	2.8	2.6	(15)	
	PrecMax	54.8	12.6	11.4	5.4	11.4	7.6	6.4	8.2	9.9	6.5	7.2	6.2	7.1	(16)	
	PrecMin	14.3	0.3	0.3	0.0	0.2	0.2	0.1	0.6	0.2	1.0	0.0	0.3	0.0	(17)	
	PrecStd	8.5	2.8	2.6	1.5	2.2	1.6	1.5	1.9	2.0	1.4	1.7	1.5	1.9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.4	92.8	84.9	77.7	65.6	58.2	57.1	61.6	70.9	78.4	87.7	90.7	(19)	
		MCWB	63.4	65.7	60.2	58.3	53.8	49.3	47.6	50.0	54.1	57.2	60.5	61.2	(20)	
	2%	DB	90.3	87.1	80.7	72.3	62.3	55.3	54.0	58.0	66.6	74.0	82.3	85.2	(21)	
		MCWB	63.3	63.9	60.2	56.5	52.0	48.4	46.7	47.5	52.6	55.7	57.6	60.5	(22)	
	5%	DB	86.8	82.6	76.9	68.8	59.6	53.0	51.6	55.1	63.2	70.4	77.7	81.8	(23)	
		MCWB	63.0	62.4	59.9	55.4	51.1	47.8	45.9	46.8	51.5	54.9	58.2	60.0	(24)	
	10%	DB	83.1	78.9	73.3	65.5	57.1	51.3	49.4	52.4	60.1	66.9	73.4	78.1	(25)	
		MCWB	62.4	62.0	59.1	54.6	50.4	47.4	44.4	45.6	50.0	53.9	57.5	59.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.3	70.1	67.6	61.9	57.5	52.8	52.9	53.3	57.8	61.5	65.5	67.3	(27)	
		MCDB	82.1	82.3	74.5	69.4	60.7	54.7	54.6	57.6	65.6	69.3	76.4	77.9	(28)	
	2%	WB	67.5	67.8	64.7	59.5	54.7	50.8	49.3	50.3	55.4	59.4	63.3	65.3	(29)	
		MCDB	80.3	78.5	71.5	67.1	59.3	52.9	51.6	55.0	62.2	68.1	74.1	75.3	(30)	
	5%	WB	66.1	66.0	63.1	57.6	52.5	49.1	46.8	48.2	53.3	57.5	61.5	63.5	(31)	
		MCDB	79.3	75.2	70.3	65.0	57.8	52.1	49.9	53.4	60.0	65.4	71.7	73.9	(32)	
	10%	WB	64.8	64.3	61.4	56.1	50.7	47.5	45.0	46.1	51.4	55.6	59.5	61.8	(33)	
		MCDB	76.9	73.2	69.3	63.3	55.7	50.6	48.3	51.4	57.6	64.1	68.5	72.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	23.8	21.7	21.0	20.9	18.3	14.4	14.3	16.4	19.0	20.8	22.2	23.5	(35)	
		MCDBR	29.0	27.1	26.4	25.4	22.9	18.8	19.0	22.1	23.7	26.5	28.9	28.7	(36)	
	5% WB	MCWBR	9.9	9.8	11.2	12.2	14.1	12.9	12.6	13.9	13.1	13.3	12.1	10.9	(37)	
		MCWBR	23.3	21.2	19.6	20.2	18.5	15.1	14.6	18.0	19.5	21.3	22.4	22.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.323	0.323	0.302	0.284	0.271	0.263	0.259	0.266	0.292	0.305	0.321	0.315	(40)	
		taud	2.549	2.577	2.617	2.637	2.622	2.667	2.666	2.630	2.532	2.521	2.516	2.563	(41)	
	Ebn at Noon		322	316	312	301	289	285	292	304	310	318	321	326	(42)	
		Edh at Noon	35	32	29	25	23	20	21	25	31	34	36	34	(43)	
(44) All-Sky Solar Radiation	RadAvg		2351	2030	1712	1353	1002	760	864	1152	1600	1973	2162	2381	(44)	
	RadStd		135	168	145	108	67	66	61	86	124	169	200	137	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.92	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)		+0.79	N/A	N/A	+1.53	N/A	N/A	-4	-153	+333	+176

Nomenclature: See separate page